



Universidad del Desarrollo

Class 11: Unsupervised Learning: Clustering Analysis

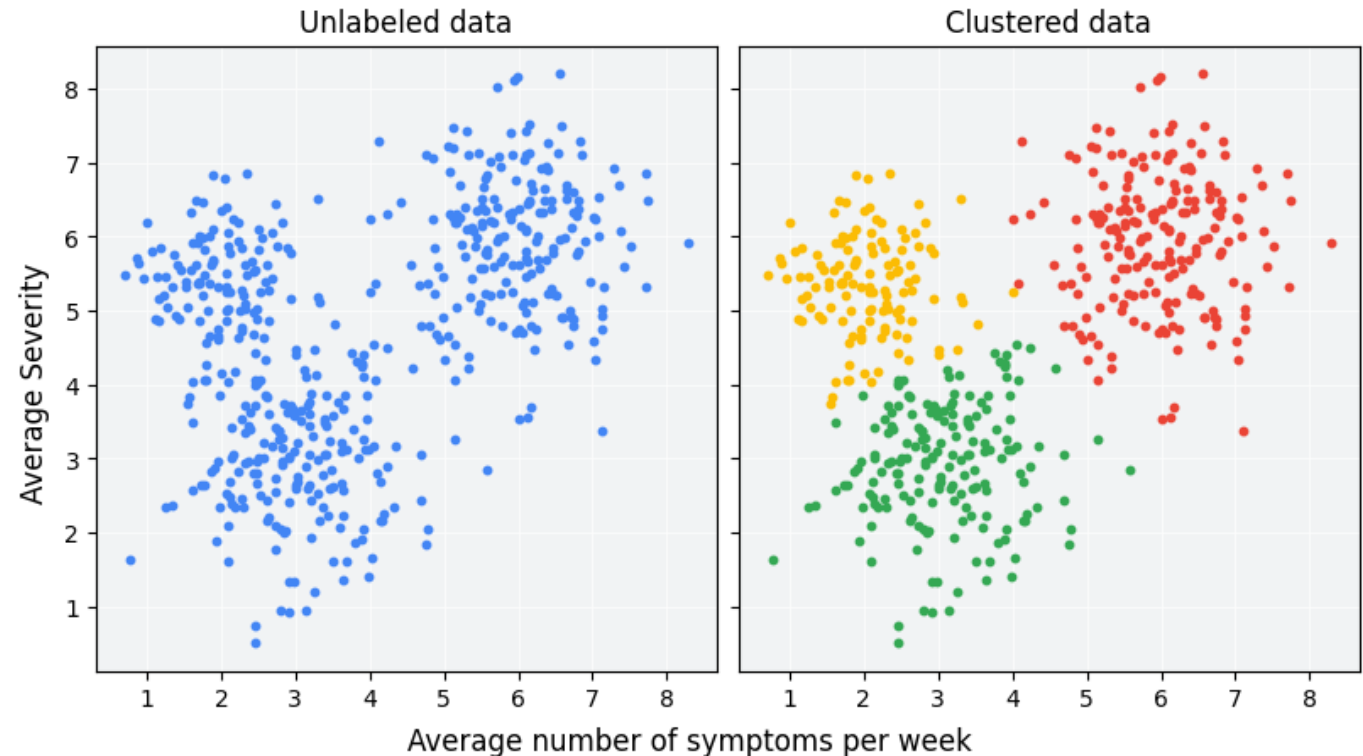
UNIT III: Advanced Analytics and Machine Learning Models

Sebastián Muñoz-Herrera, PhD.
Héctor Hormazábal

Ingeniería
UDD

Unsupervised Learning - Foundations of Cluster Analysis

- Transitioning from predictive models **with explicit target guidance** to the unsupervised partition of raw data matrices without historical training labels.
- Implementing the Euclidean distance metric $d(x, y) = \|x - y\|$ to establish a standardized **measure of geometric proximity** between high-dimensional observations.
- **Minimizing internal distance variance** to transform a continuous feature space into discrete, homogeneous regions with defined boundaries.
- Initializing the K-Means algorithm as the specific optimization engine that will locate stable cluster centers to maximize the compactness of each group.



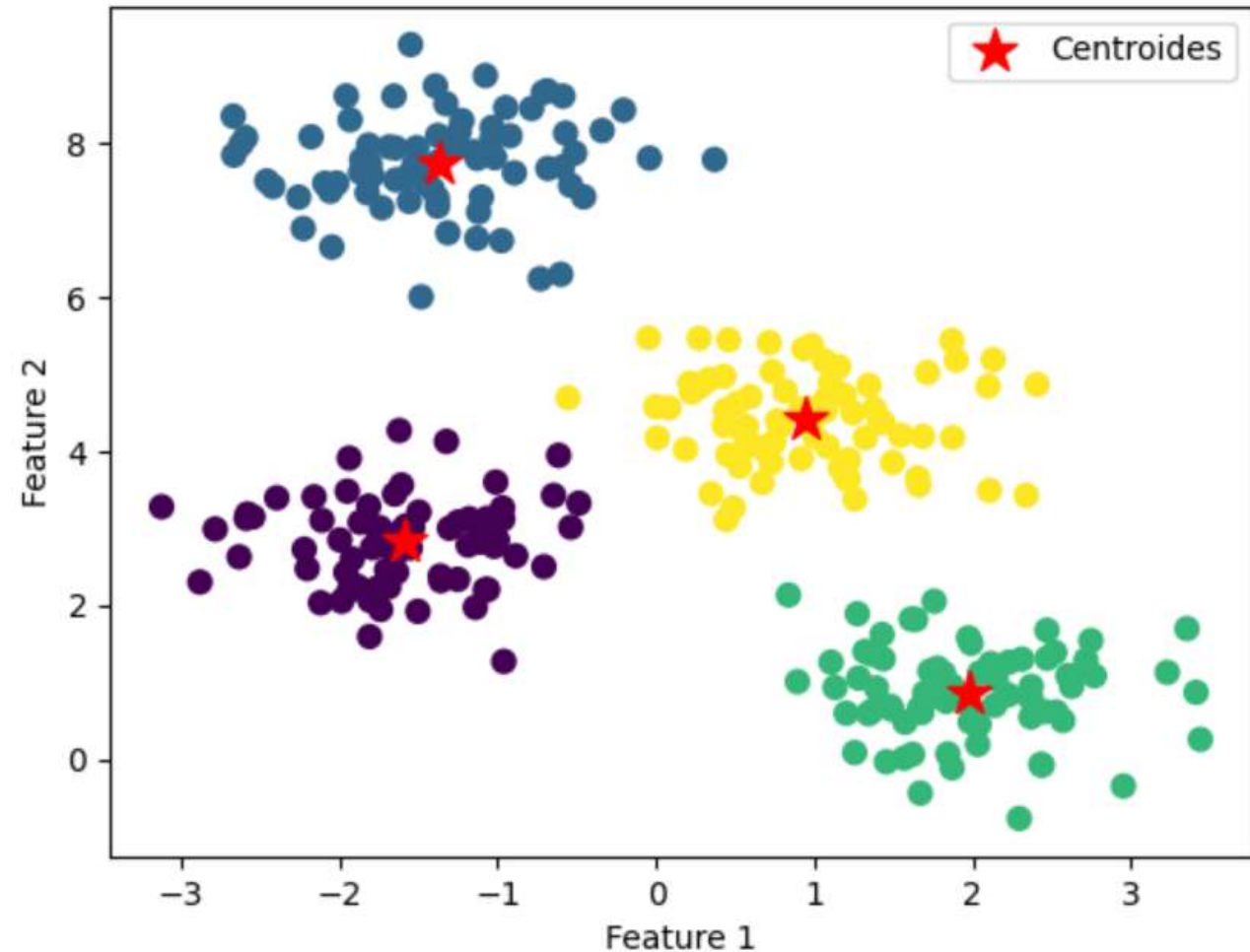
The Core Intuition of *K*-Means – Clustering ML method

- Objective-driven **partitioning that seeks to split a dataset into *k* distinct clusters**, ensuring that observations mapped to the same subset S_j , $j = \overline{1, k}$ share maximum behavioral similarity.

- Relying on the concept of centroids μ_j , which act as the mathematical gravity **centers or mean vectors** that represent the coordinates of all elements assigned to that specific group.

$$\mu_j = \frac{1}{|S_j|} \sum_{x_i \in S_j} x_i$$

- Minimizing the intra-cluster variance by finding stable positions for these centers, creating a geometric state where no individual observation can switch groups without increasing the global error.



The K -Means Iterative Optimization

- The **input dataset is a matrix X containing n observations and p features**, where each individual observation: $x_i = [x_{\{i1\}}, x_{\{i2\}}, \dots, x_{\{ip\}}]$ is a vector in the p -dimensional real space $\mathbb{R}^p$.
- The objective is to split these n observations into k disjoint subsets $S = \{S_1, S_2, \dots, S_k\}$, representing the clusters, based on their spatial proximity.
- The optimization handles the minimization of the objective function J (Inertia), which sums the **squared Euclidean distances** between each data vector x_i and the centroid μ_j of its assigned cluster S_j .

$$J = \sum_{j=1}^k \sum_{x_i \in S_j} \|x_i - \mu_j\|^2$$

- The mathematical partition enforces that every observation belongs exclusively to one group ($S_j \cap S_l = \emptyset, j \neq l$) meaning assignments are discrete and non-probabilistic.
- The process triggers by setting the initial state of the cluster centers (randomly) $\mu_1^{\{(0)\}}, \dots, \mu_k^{\{(0)\}}$ using coordinates from the dataset, which establishes the first geometric boundaries.

The Iterative Assignment and Update Cycle

- The Assignment step (*often called the Expectation or E-step*) keeps the centroids μ_j fixed while each observation x_i joins the group of its nearest center, **minimizing the squared Euclidean distance**:

$$\operatorname{argmin}_j \|x_i - \mu_j\|^2$$

- The Update step (*often called the Maximization or M-step*) keeps the cluster assignments fixed while every centroid μ_j migrates to the arithmetic mean (geometric center of mass) of its current members:

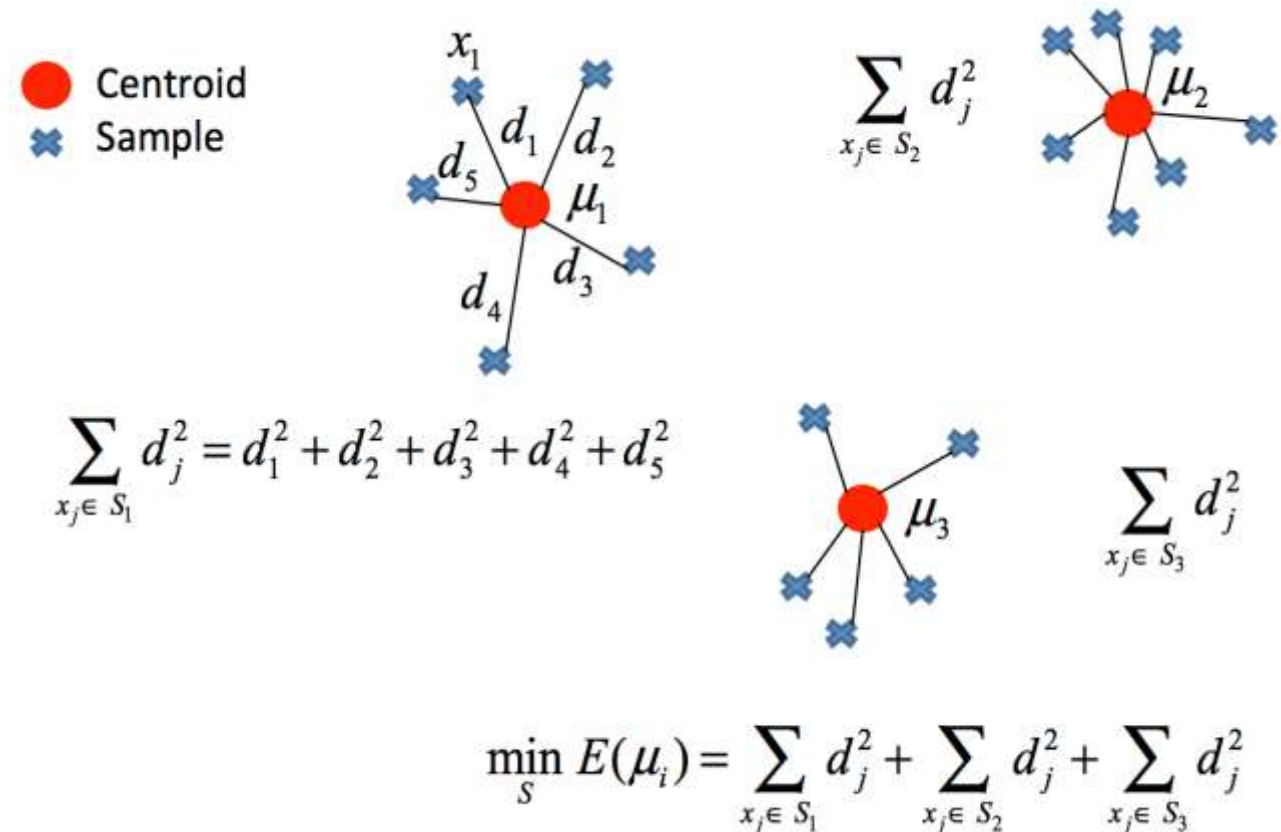
$$\mu_j = \frac{1}{|S_j|} \sum_{x_i \in S_j} x_i$$

TERMINATION CRITERION:

- The algorithm cycles through these two steps recursively **until reaching a state of stability where no observations change cluster membership**, indicating that the system has converged to a local optimum.
- Practical termination also occurs if the process reaches a **maximum number of iterations or if the physical displacement of the centroids between steps becomes smaller than a set tolerance threshold ϵ** .

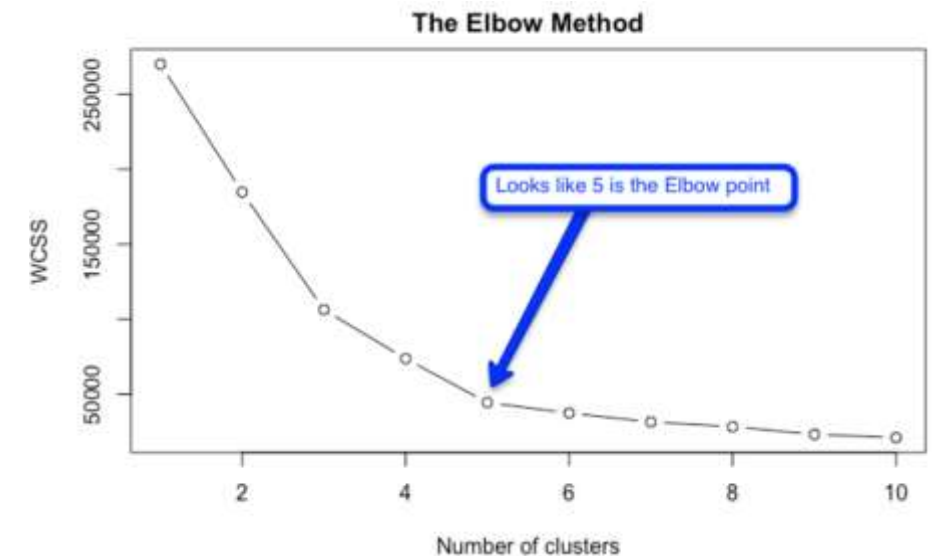
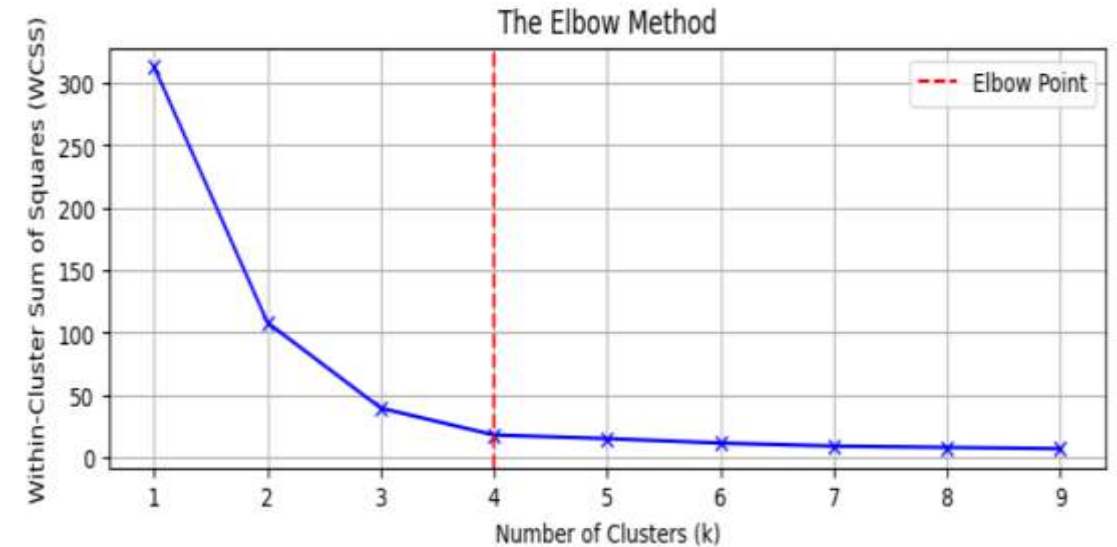
K-means. Initialization Sensitivity

- Simultaneous optimization of discrete assignments and centroid coordinates results in a **non-convex error surface** with multiple local minima.
- Random seeding limits the iterative trajectory to a local search space, often **missing the global optimal partition**.
- Poorly placed initial centers cause high-density groups to be artificially split or merged.
- To avoid these poor initial configurations, the algorithm can either be initialized strategically via **K-Means++ (heuristic)** or run multiple independent times.



Optimal K Selection: The Elbow Method

- The total Inertia $J(k)$ is a **monotonically non-increasing function of k** that naturally scales down toward zero as the number of clusters approaches the total number of observations (n).
- The optimal number of clusters is identified at the "elbow" point, which represents the maximum curvature where the second-order difference in the objective function is maximized.
- This selection criteria formalizes the marginal reduction in variance $\Delta J = \Delta J(k + 1) - \Delta J(k)$ capturing the precise point of diminishing returns.
- Beyond this geometric threshold, any additional cluster increases model complexity and risks over-fitting without providing a significant gain in explained variance.



Validation via Silhouette Analysis

- The **Silhouette Coefficient** evaluates how well-defined each cluster is by measuring the trade-off between inner compactness and outer separation.
- **Intra-cluster distance** (a_i): The average distance between the observation x_i and all other points in its same cluster. It quantifies **cohesion** (lower means - tighter clusters).
- **Nearest-cluster distance** (b_i): The average distance between the observation x_i and all points in the closest neighboring cluster. It quantifies **separation** (higher means more isolated clusters).
- The individual silhouette score normalizes these components to evaluate the quality of the assignment:

$$s_i = \frac{b(i) - a(i)}{\max(a_i, b_i)}$$

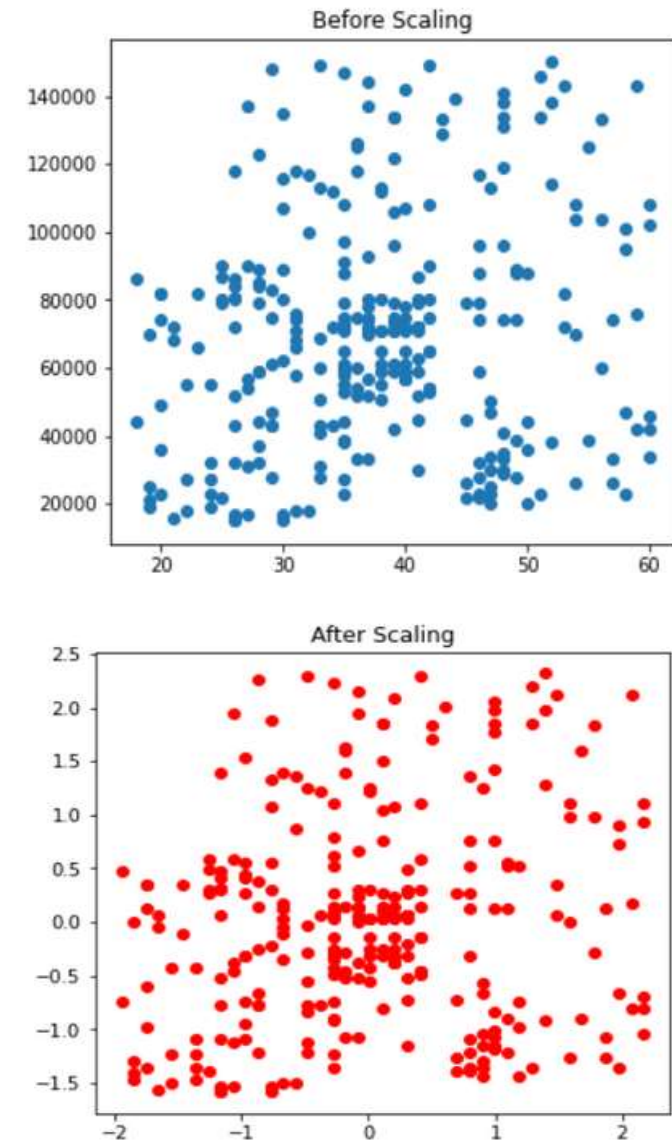
- The metric ranges from -1 to 1, where values near 1 indicate highly distinct segments, 0 indicates overlapping boundaries, and negative values reveal misallocated points.

Critical Pre-processing: Data Scaling - Z score

- Because K -Means relies strictly on the **Euclidean distance**, features with larger numerical ranges will mathematically dominate the distance calculations.
- Without proper normalization, **variables with high variance artificially dictate cluster boundaries**, rendering lower-scale variables geometrically irrelevant.
- Standardizing the features (Z-score scaling) transforms the dataset so that each variable has a mean of 0 and a variance of 1:

$$z_i = \frac{x_i - \mu}{\sigma}, i = \overline{1, n}$$

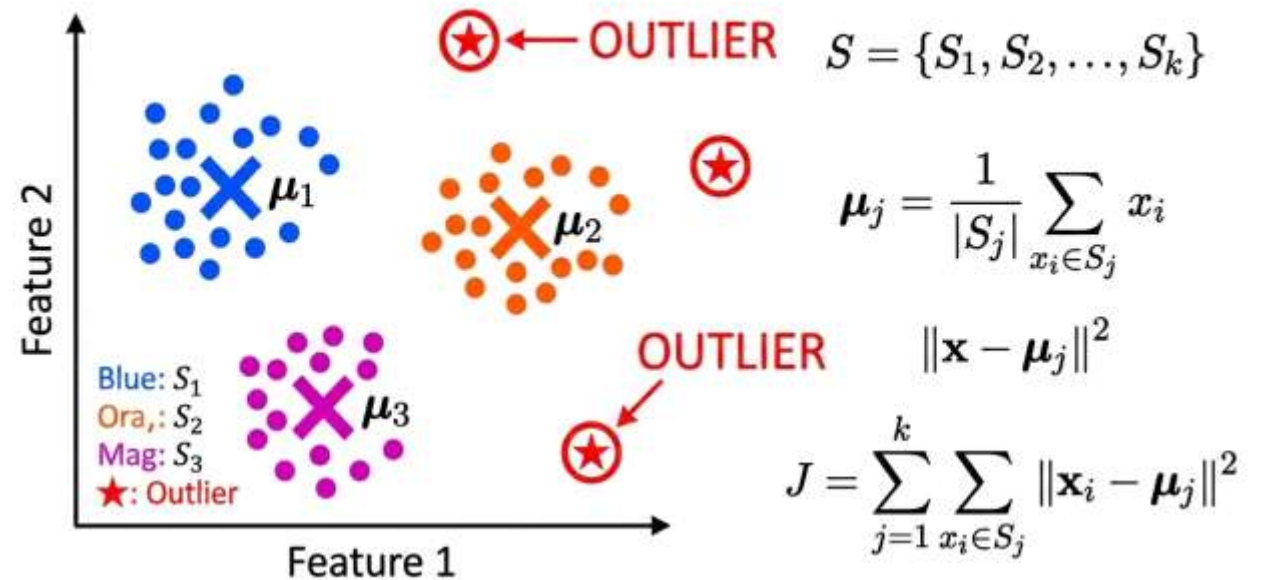
- Feature scaling ensures an isotropic feature space where all dimensions contribute equitably to the calculation of centroids and cluster updates.



Algorithm Limitations and Geometric Assumptions

- The arithmetic mean update mechanism makes the algorithm **highly sensitive to outliers**, which pull centroids away from the high-density core of the clusters.
- K-Means **implicitly assumes clusters are spherical and have a uniform spread in all directions**; it fails to capture complex geometries like moons, rings, or elongated shapes.
- The algorithm struggles to correctly partition groups with significantly different densities or imbalanced membership sizes, often causing artificial splits in larger clusters.
- Defining k upfront remains a fundamental constraint, as the model lacks an internal mechanism to automatically discover the natural number of groups.

K-Means Clustering: Sensitivity to Outliers



K-Means Inference and Prediction Architecture

- **New observations must be standardized** using the exact parameters (mean μ and standard deviation σ) derived from the training dataset to ensure spatial consistency.
- The model evaluates the newly normalized vector $x_{\{new\}}$ against the **frozen final coordinates of the k centroids** using:

$$d(x_{new}, \mu_j) = \|x_{new} - \mu_j\|$$

- Classification is executed as a deterministic **argmin** operation, mapping the new observation to the single closest cluster index:

$$\text{Cluster}(x_{new}) = \operatorname{argmin}_j \|x_{new} - \mu_j\|^2$$

- Fast vector-distance calculation enables real-time inference, allowing live data pipelines to instantly flag operational risk or assign customer personas during live sessions.

Practical Section



Google Colab — RFM Clustering
K means ML algorithm
Cluster generation + Business impact
Evaluation metrics and final centroids
Visualization – Elbow method



Universidad del Desarrollo

Class 11: Unsupervised Learning: Clustering Analysis

UNIT III: Advanced Analytics and Machine Learning Models

Sebastián Muñoz-Herrera, PhD.
Héctor Hormazábal

Ingeniería
UDD